

8086 Flag Control & Compare Instructions

1. Flag Control Instructions

Concept

Flag control instructions modify specific bits of the **FLAGS register**. They control the internal behavior of the processor such as carry handling, interrupts, and direction of string operations.

1.1 LAHF (Load AH from Flags)

Concept

LAHF transfers the lower byte of the FLAGS register into the AH register.

Rules / Notes

Bit Mapping (LSB → MSB):

- Bit 0: Carry Flag (CF)
- Bit 2: Parity Flag (PF)
- Bit 4: Auxiliary Carry (AF)
- Bit 6: Zero Flag (ZF)
- Bit 7: Sign Flag (SF)

Other bits are undefined.

Example

```
LAHF      ; AH <- low byte of FLAGS
```

Exam Tip

LAHF is useful for saving flags before modifying them.

1.2 SAHF (Store AH into Flags)

Concept

SAHF loads selected flag bits from AH into the FLAGS register.

Rules / Notes

Flags affected:

SF, ZF, AF, PF, CF

Other flags remain unchanged.

Example

```
MOV AH, 40H    ; set ZF
SAHF
```

Exam Tip

SAHF restores flags previously saved using LAHF.

1.3 Carry Flag Control Instructions

CMC (Complement Carry)

Concept

CMC toggles the Carry Flag.

$$CF \leftarrow \overline{CF}$$

Example

```
CMC    ; CF is inverted
```

CLC (Clear Carry)

Concept

CLC clears the Carry Flag.

$$CF \leftarrow 0$$

Example

```
CLC
```

STC (Set Carry)

Concept

STC sets the Carry Flag.

$$CF \leftarrow 1$$

Example

STC

Exam Tip

CMC, CLC, and STC affect only CF.

1.4 Interrupt Flag Control

CLI (Clear Interrupt)

Concept

CLI clears the Interrupt Flag. External maskable interrupts are disabled.

$$IF \leftarrow 0$$

Example

CLI

STI (Set Interrupt)

Concept

STI sets the Interrupt Flag. External interrupts are enabled.

$$IF \leftarrow 1$$

Example

STI

Exam Tip

CLI/STI are critical in operating systems and interrupt handling.

1.5 Direction Flag Control

CLD (Clear Direction)

Concept

CLD clears the Direction Flag. String instructions increment index registers.

$$DF \leftarrow 0$$

Example

CLD

STD (Set Direction)

Concept

STD sets the Direction Flag. String instructions decrement index registers.

$$DF \leftarrow 1$$

Example

STD

2. CMP (Compare) Instruction

Concept

CMP compares two operands by performing a subtraction. The result is discarded, but flags are updated.

Rules / Notes

Key Characteristics

- Compares two 8-bit or two 16-bit operands
- Operands may be register, memory, or immediate
- Uses 2's complement subtraction
- Result is not stored
- Updates: CF, AF, OF, PF, SF, ZF

Operation

$$\text{CMP } D, S \Rightarrow D - S$$

- If S is immediate, it is sign-extended
- Only flags reflect the comparison result

Example

CMP AX, BX

Meaning: Compare AX with BX (AX - BX).

Example

```
CMP AL, 05H
```

Meaning: Compare AL with 5.

Example

```
CMP [2000H], AX
```

Meaning: Compare memory word at DS:2000H with AX.

Exam Tip

CMP behaves like SUB, but does not store the result. Only flags matter.

Quick Revision (One Look)

- **LAHF/SAHF:** transfer flags to/from AH
- **CLC/STC/CMC:** carry flag control
- **CLI/STI:** interrupt enable/disable
- **CLD/STD:** string direction control
- **CMP:** subtraction-based comparison (flags only)